THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Easy level: direct, single-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The HCF of 12 and 18 is: 1

(a) 12 **(b)** 3 **(c)** 6 **(d)** 36

2. If $p(x) = x^2 - 3x + 2$, then $p(4)$ is: 1

(a) 4 **(b)** 30 **(c)** 8 **(d)** 6

3. The pair of equations $x + 2y = 3$ and $2x + 4y = 6$ has: 1

(a) no solution (parallel lines)
(b) exactly two solutions
(c) infinitely many solutions (coincident lines)
(d) exactly one solution (intersecting lines)

4. Which of the following is a quadratic equation? 1

(a) $3x + 2 = 0$ **(b)** $x + 3 = 2$ **(c)** $x^2 + 3x + 2 = 0$ **(d)** $x^3 + 3x + 2 = 0$

5. The common difference of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) -4 **(b)** 4 **(c)** 5 **(d)** 8

6. In a right triangle with legs 3 and 4, the hypotenuse is: 1

(a) 1 **(b)** 7 **(c)** 5 **(d)** 25

7. The distance of the point $(3, 4)$ from the origin is: 1

(a) 25 **(b)** 7 **(c)** 5 **(d)** 1

8. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\sin\theta =$ 1

(a) $\frac{4}{3}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{4}{5}$

9. From a point on level ground, 7 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. The height of the tower is: 1

(a) 7 m **(b)** $\frac{7\sqrt{3}}{3}$ m **(c)** $7\sqrt{3}$ m **(d)** 14 m

10. A point P is 5 cm from the centre O of a circle of radius 3 cm. The length of the tangent drawn from P to the circle is: 1

(a) 4 cm **(b)** 16 cm **(c)** 8 cm **(d)** 2 cm

11. The radius of a circle is 7 cm. The area of the circle is: 1

(a) 44 cm² **(b)** 308 cm² **(c)** 22 cm² **(d)** 154 cm²

12. A right circular cylinder has base radius 7 cm and height 5 cm. The curved surface area of the cylinder is: 1

(a) 440 cm² **(b)** 528 cm² **(c)** 220 cm² **(d)** 770 cm²

13. The class interval is 10–20. The class mark is: 1

(a) 15 **(b)** 20 **(c)** 10 **(d)** 16

14. A die is thrown once. The probability of getting an even number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{2}$

15. The LCM of 4 and 6 is: 1

(a) 2 **(b)** 24 **(c)** 12 **(d)** 10

16. The zero of the linear polynomial $2x - 8$ is: 1

(a) -4 **(b)** 8 **(c)** 4 **(d)** -8

17. Which of the following pairs (x, y) is a solution of $2x + 3y = 12$? 1

(a) $(-3, 2)$ **(b)** $(3, -2)$ **(c)** $(2, 3)$ **(d)** $(3, 2)$

18. Which of the following is a root of $x^2 - 5x + 6 = 0$? 1

(a) -2 **(b)** 2 **(c)** 5 **(d)** 1

19. The 10th term of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 40 **(b)** 39 **(c)** 35 **(d)** 43

20. The ratio of corresponding sides of two similar triangles is 1:3. Their perimeters are in the ratio 1: 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** 9

21. The midpoint of the join of $(2,\ 4)$ and $(6,\ 8)$ is: 1

(a) $(8,\ 12)$ **(b)** $(4,\ 7)$ **(c)** $(4,\ 6)$ **(d)** $(5,\ 6)$

22. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\tan\theta =$ 1

(a) $\frac{4}{3}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{3}{5}$

23. From a point on level ground, 5 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) 5 m **(b)** $10\sqrt{3}$ m **(c)** $5\sqrt{3}$ m **(d)** $\frac{5\sqrt{3}}{3}$ m

24. The length of the tangent from an external point P to a circle is 8 cm and P is 10 cm from the centre. The radius of the circle is: 1

(a) 18 cm **(b)** 9 cm **(c)** 2 cm **(d)** 6 cm

25. The radius of a circle is 7 cm. The circumference of the circle is: 1

(a) 44 cm **(b)** 88 cm **(c)** 154 cm **(d)** 22 cm

26. A right circular cylinder has base radius 7 cm and height 10 cm. The volume of the cylinder is: 1

(a) 1540 cm³ **(b)** 748 cm³ **(c)** 3080 cm³ **(d)** 440 cm³

27. For the data with classes 10–20, 20–30, 30–40, 40–50 with frequencies 5, 8, 20, 17 respectively, The modal class is: 1

(a) 20–30 **(b)** 10–20 **(c)** 30–40 **(d)** 40–50

28. A die is thrown once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

29. The number of prime factors (counted with repetition) of 12 is: 1

(a) 2 **(b)** 6 **(c)** 3 **(d)** 4

30. The sum of the zeroes of $x^2 - 5x + 6$ is: 1

(a) 11 **(b)** 5 **(c)** 6 **(d)** -5

31. If $x = 2, y = 1$ is a solution of $2x + 3y = k$, then k is: 1

(a) 7 **(b)** 1 **(c)** 15 **(d)** 8

32. In $2x^2 - 3x + 5 = 0$, the value of b is: 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** -3

33. The sum of the first 10 terms of $1, 2, 3, 4, \ldots$ is: 1

(a) 55 **(b)** 66 **(c)** 45 **(d)** 110

34. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$, $DB = 3$, $AE = 4$. Then $EC =$ 1

(a) 3 **(b)** 4 **(c)** 8 **(d)** 6

35. The point $(3, 4)$ lies in the: 1

(a) first quadrant **(b)** second quadrant **(c)** fourth quadrant **(d)** third quadrant

36. The value of $\sin 30^\circ$ is: 1

(a) $\frac{1}{2}$ **(b)** 1 **(c)** 2 **(d)** 0

37. From a point on level ground, 9 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 9 m **(b)** $3\sqrt{3}$ m **(c)** $9\sqrt{3}$ m **(d)** 3 m

38. A tangent to a circle of radius 7 cm has length 24 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 50 cm **(b)** 25 cm **(c)** 31 cm **(d)** 17 cm

39. The circumference of a circle is 44 cm. The radius of the circle is: 1

(a) 3.5 cm **(b)** 14 cm **(c)** 7 cm **(d)** 22 cm

40. A right circular cone has base radius 7 cm and height 24 cm. The slant height of the cone is: 1

(a) 23 cm **(b)** 25 cm **(c)** 15.5 cm **(d)** 31 cm

41. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 7, 17, 20, 3, 3 respectively, The median class is: 1

(a) 30–40 **(b)** 20–30 **(c)** 40–50 **(d)** 50–60

42. A coin is tossed two times. The probability of getting two heads is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{3}{4}$ **(c)** 0 **(d)** $\frac{1}{2}$

43. Which of the following is an irrational number? 1

(a) $\sqrt{7}$ **(b)** $\sqrt{25}$ **(c)** $\frac{5}{8}$ **(d)** 25

44. The product of the zeroes of $x^2 - 5x + 6$ is: 1

(a) -6 **(b)** 6 **(c)** 11 **(d)** 5

45. The sum of two numbers is 10 and their difference is 4. The larger number is: 1

(a) 6 **(b)** 7 **(c)** 14 **(d)** 3

46. The roots of $(x-2)(x-3)=0$ are: 1

(a) 5 and 6 **(b)** 2 and 3 **(c)** 2 and -3 **(d)** -2 and -3

47. Which of the following sequences is an AP? 1

(a) $4, 7, 11, 15, \ldots$ **(b)** $4, 7, 13, 16, \ldots$ **(c)** $4, 7, 13, 22, \ldots$ **(d)** $4, 7, 10, 13, \ldots$

48. The areas of two similar triangles are in the ratio 4:9. The ratio of their corresponding sides is 2: 1

(a) 9 **(b)** 5 **(c)** 6 **(d)** 3

49. The coordinates of the point on the x-axis at distance 3 from the origin on the right are: 1

(a) $(3,\ 3)$ **(b)** $(3,\ 0)$ **(c)** $(-3,\ 0)$ **(d)** $(0,\ 3)$

50. The value of $\sin^2 20^\circ + \cos^2 20^\circ$ is: 1

(a) 20 **(b)** 2 **(c)** 1 **(d)** 0

END OF QUESTION PAPER **Total: 50 Marks**

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